

B.Tech ECE - Unit 3 Full Solutions (PSLP) Exam Ready Format

Q1: t-Test (Girls vs Boys Heights)

$H_0: \mu_1 = \mu_2$, $H_1: \mu_1 > \mu_2$

Mean (Girls) = 68.17, Mean (Boys) = 67.67

Variance (Girls) ≈ 10.97 , Variance (Boys) ≈ 18.75

$t = 0.30$, $df = 13$, $t_{table} = 1.77$

Conclusion: Since $t_{cal} < t_{table} \rightarrow$ Accept H_0 . No significant difference.

Q2: Karl Pearson Correlation

x	y	x^2	y^2	xy
1	6	1	36	6
5	1	25	1	5
3	0	9	0	0
2	0	4	0	0
1	1	1	1	1
1	2	1	4	2
7	1	49	1	7
3	5	9	25	15

$r \approx -0.29 \rightarrow$ Weak negative correlation

Q3: Chi-Square Test (Poisson Fit)

Mean $\lambda = 2$

$\chi^2_{cal} \approx 3.5$, $\chi^2_{table} = 9.49$

Conclusion: $\chi^2_{cal} < \chi^2_{table} \rightarrow$ Accept $H_0 \rightarrow$ Data follows Poisson distribution

Q4: F-Test

$s_1^2 = 10$, $s_2^2 = 9.82$

$F = 1.02$, $F_{table} = 2.90$

Conclusion: Accept $H_0 \rightarrow$ Variances are equal